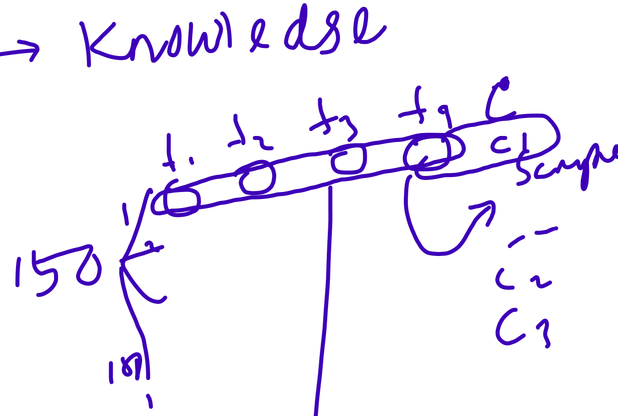
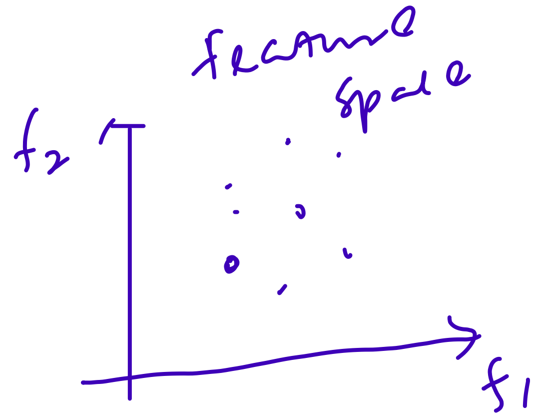
Data → Data Preprocessing → Data Mining → Knowledge



Data

Attribute/feature

Sample/Object/instance/entity/case/record point



$[f_1, f_2]$

$f = [f_1, f_2, f_3, f_4]$

↓ feature vector

dimension is 4

Types of attributes:

Nominal:

Ordinal: 1-5.

Interval: value that represents interval

Ratio: free no.
→ discrete & continuous

Person id	Refund	Marital Status	Taxable Income	Cheat Label
1	yes	single	125k	No
2	NO	Married	70k	No
3	No	Divorced	60k	Yes

Operations: (1) Distinctness = ≠
(2) Order < >
(3) Addition + -
(4) Multiplication * /

Discrete attribute: Integer

Special case: 0 & 1

Binary attribute

* Continuous Attribute :

KNN

f_1	f_2	f_3	f_4
0	0	0	0
1	1	1	1

(0-1)

$\sigma(Wx+b) \rightarrow (0,1)$

$n\times d \quad d\times 1 \quad n\times 1$

NN

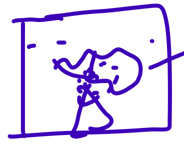
MNIST

28×28

0 9

Digit

1×784



2D

reshape

512×512

$1\times(512\times 512)$

1D

$2^9 \times 2^9 = 2^{18}$

Types of the dataset:

① Record
- data matrix → Iris $N \times d$
  - d → no. of features
  - N → no. of samples
- document data → Ex: Sports news article
- transaction data

② Graph

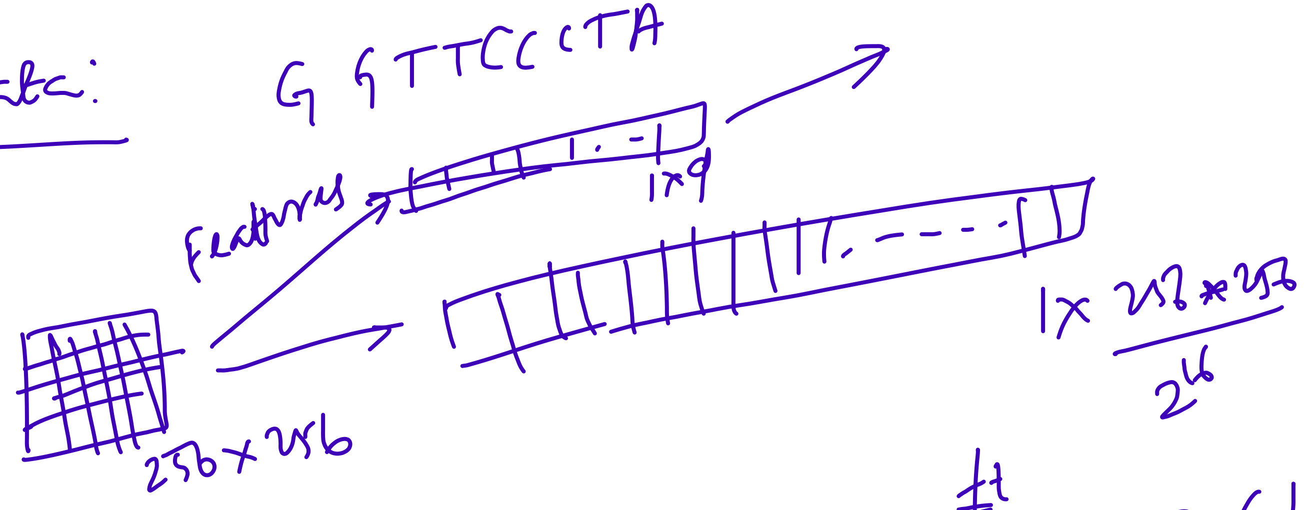
③ Ordered

	Team	Coach	Play	Ball	Score	Game	Win	Loss	Timeout	Season
Doc 1	3	0	5	2	0	6	0	2	1	1
Doc 2										
⋮										

TransID	Items
1	Bread, Coke, Milk
2	Butter, bread, Diapers
3	Coke, Biscuits, Milk

Bread	Coke	Milk	Biscuits	Diapers
1	1	1	0	0

Ordered Data:



#t

$6.11 \rightarrow 611$

Data Quality:

Problems: - Noise & Outliers ✓
- Missing values
- Duplicate data

Data Cleaning

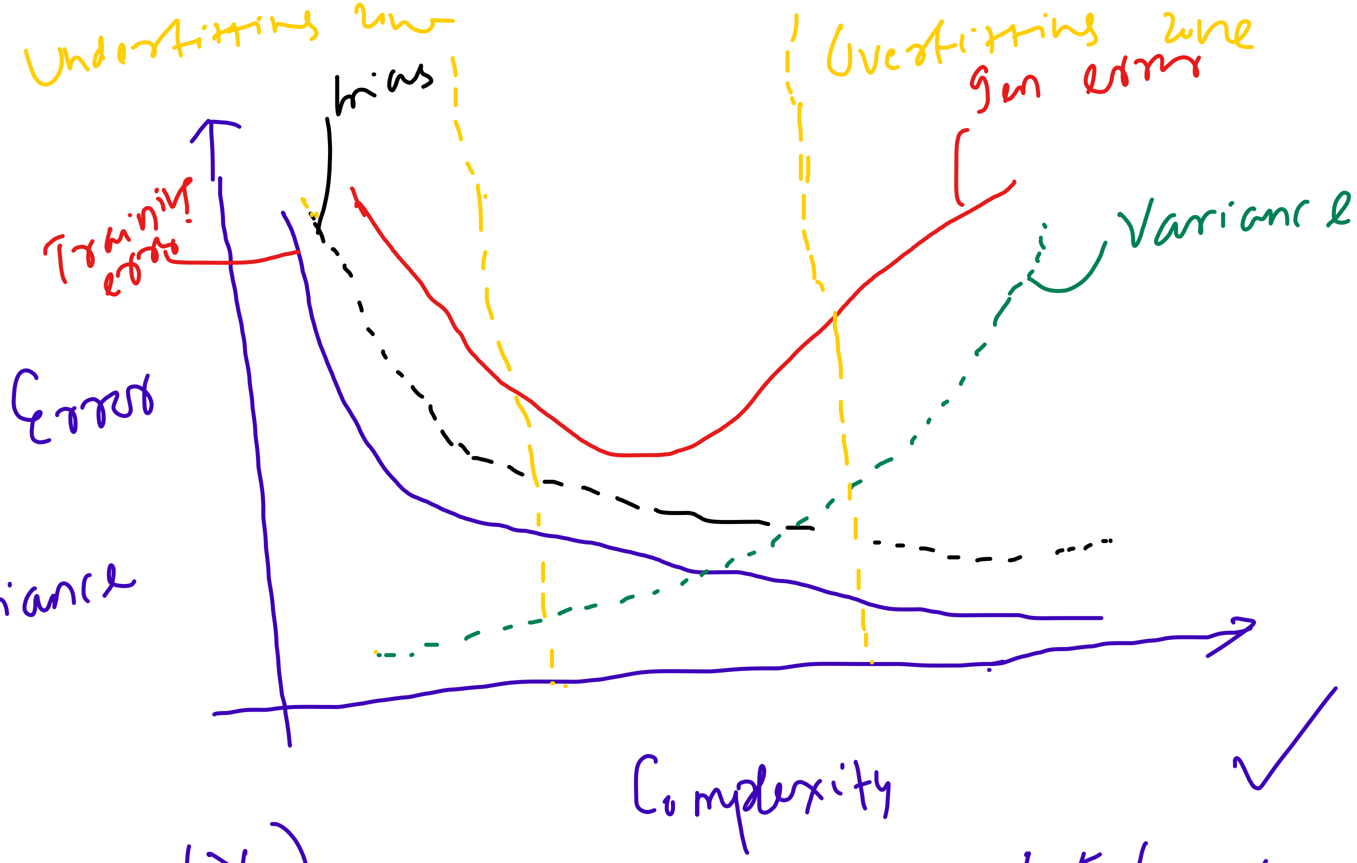
X x_1, x_2, ..., x_n

f_1 f_2 f_3 f_4

mean/median

mode

Bias Variance Tradeoff:



(1)
Train error: 1%
Validation error: 12%
(Overfitting)
Low bias & high variance

(2)
15%
17%
(underfitting)
High bias
&
low variance

(3) (X)
15%
30%
High bias
&
high Variance

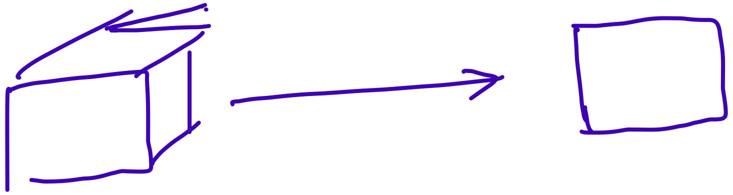
(4)
0.5%
1%
(proper fit)
low bias
&
low variance ✓

$$\text{Bias}^2 = E\{[\hat{y} - y]^2\}$$

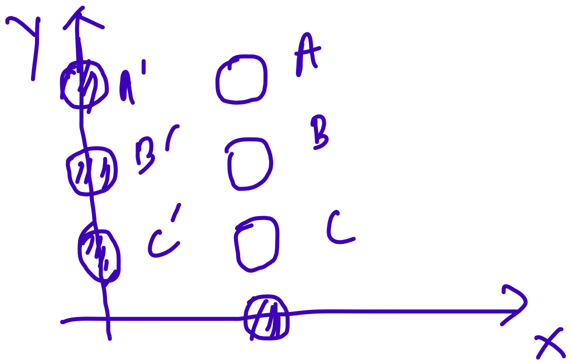
$$\text{Var} = E\{[\hat{y} - \bar{\hat{y}}]^2\}$$

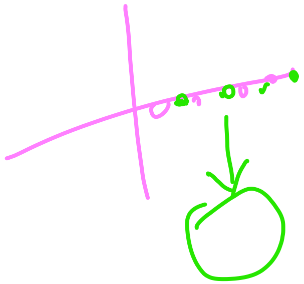
Principal Component Analysis (PCA):

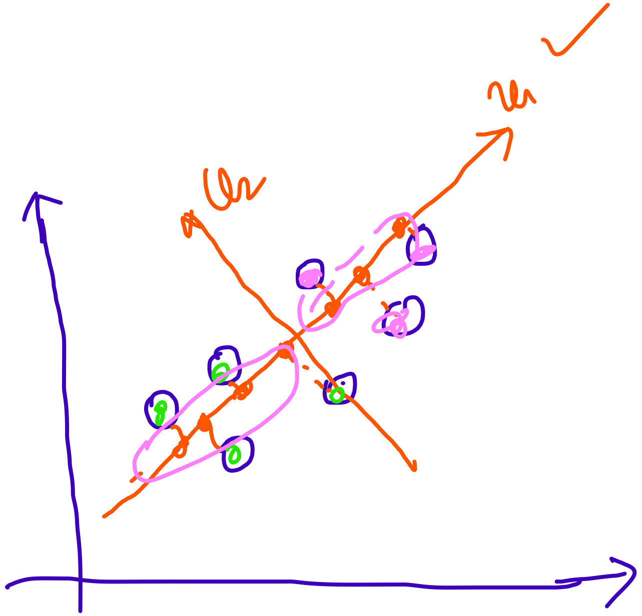
3D → 2D

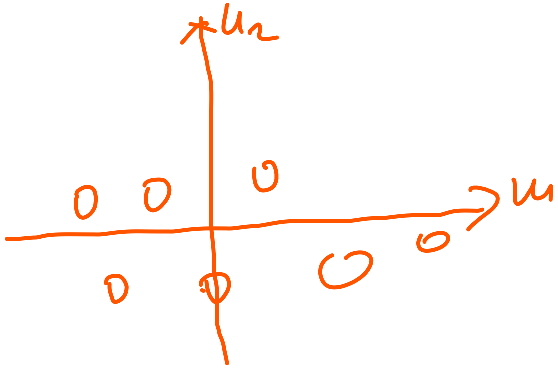


2D → 1D









Essence of PCA:

- We project our high-dimensional data to a new coordinate system
  ↳ axes are orthogonal to each other
- Choose only a few of these dimensions that explains most of the variations in the data

PCA:

Data $X = \begin{bmatrix} & \end{bmatrix}_{n \times d}$ where n = # samples, d = # features

$\Downarrow$ PCA

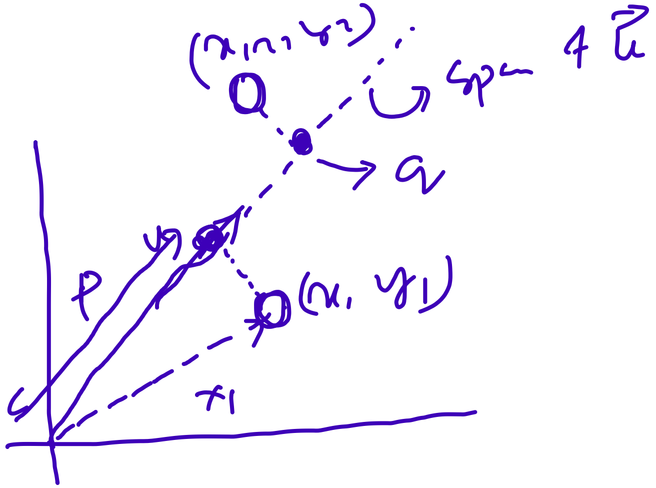
$X' = Y = \begin{bmatrix} & \end{bmatrix}_{n \times d'}$ where $d' << d$

Goal:

Project the data X on a new axis vector u such that

- The variance of the projected data is maximum ✓
- u is an unit vector ✓

Projection of data on a vector:



$X = \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \end{bmatrix}_{2\times 2}$ 2 samples, 2 features

$\|u\| = 1$

$p = \text{Proj}_u x_1 = \dfrac{x_1 . u}{\cancel{\|u\|} \rightarrow 1} = x_1 . u$ ↳ dot product

$x_1 = (x_1, y_1)$

let's $u = \begin{bmatrix} m \\ n \end{bmatrix}_{2\times 1}$ $p = x_1 . u = m x_1 + n y_1$ ✓

$X = \begin{bmatrix} \cancel{\leftarrow x_1 \rightarrow} \\ \cancel{\leftarrow x_2 \rightarrow} \\ \\ \end{bmatrix}_{n\times d}$ $u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_d \end{bmatrix}_{d\times 1}$

$\text{proj}_u X = Xu = \begin{bmatrix} x_1' \\ x_2' \\ \vdots \\ x_n' \end{bmatrix}_{n\times 1}$

$x_{11} u_1 + x_{12} u_2 + \dots + x_{1d} u_d$

PCA:

Objective function: $\arg\max_{u} [Var(Xu)]$

Constrain $\|u\| = 1$

$var(Xu) = u^T \Sigma u$

↳ Covariance matrix $d \times d$ → square & symmetric

PSD → eigenvalues are nonnegative

* eigenvectors are orthogonal

$\arg\max_{u} [u^T \Sigma u]$

$\| u \| = 1$

* Lagrangian Multiplier

SVM

$L = f(x) + \lambda g(x)$

$\frac{\partial L}{\partial x} = 0$

$\frac{\partial L}{\partial \lambda} =$

eigenvalue

$\boxed{A\vec{v} = \lambda \vec{v}}$

eigenvector

Sol^n:

$$\boxed{\Sigma u = \lambda u}$$

u

EV of → $\Sigma_{d \times d}$

step 1

$x_c = (X - \mu)$ (centered data)

$X_{n\times d} \longrightarrow \Sigma_{d\times d}$ symmetric $\longrightarrow$ Eigen values & eigen vectors

$\lambda_1 > \lambda_2 > \dots > \lambda_d$ ↳ defines the variance

$u_1 \ u_2 \ \dots \ u_d$

$U_{d\times d}$

U

$Y = X_c U = (n\times d) \times (d\times d) = (n\times d)$

$(n\times d) \times (d\times 1) = n\times 1$

PCs (Principal Components)

PCA

How to choose d' ?

$$1^{st}\ PC\ (u_1) = \frac{\lambda_1}{\lambda_1 + \lambda_2 + \dots + \lambda_d} \times 100\ \%$$

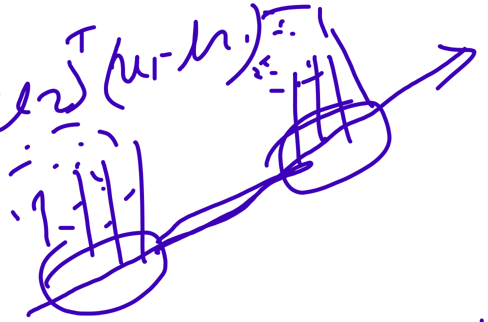
$$\text{two PCs} = \frac{\lambda_1 + \lambda_2}{\lambda_1 + \lambda_2 + \dots + \lambda_d}$$

$$\frac{\sum_{i=1}^{d'} \lambda_i}{\sum_{j=1}^{d} \lambda_j} > 0.9$$

Linear Discriminant Analysis (LDA)

$$\boxed{S_W^{-1} S_B V = \lambda V}$$

$S_B \to (\mu_1-\mu_2)^T(\mu_1-\mu_2)$



S_W $\quad$ S_B

$d\times d$ $\quad$ $d\times d$

within scatter

$S_1 + S_2$

$N \times d$

$\Sigma \; d\times d$

Data Preprocessing:

$x_1, x_2, \& x_3 \longrightarrow X$ unifind

Data integration: Combines the data from multiple sources

RF & Sampling: ~~Size~~ Selecting a subset of datapoints from the dataset

↳ Random sampling

Aggregation:

Data normalization: -1 to 1, 0 to 1

Data reduction:

Discretization & binarization:

f_1	f_2
1W0	1
|	
1	10

Data Normalization:

$[0, 255] \rightarrow [0, 1]$

✓ Max-min normalization: $z = \frac{X - X_{min}}{X_{max} - X_{min}}$

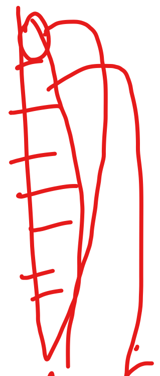
BN *✓ Z-score normalization: $z = \frac{X - \mu}{\sigma}$

Logarithmic transformation

Root transform

Decimal scaling

f_1



μ, σ

zero mean

$1 \rightarrow \sigma$

$$Z = \frac{x - x_{min}}{x_{max} - x_{min}} \left(x_{max}^{new} - x_{min}^{new} \right) + x_{min}^{new}$$

Data Reduction → Dimensionality Reduction (PCA, LDA) → $x_u \to y'$

Data Reduction → Feature selection

1 2 3 4 .. 8 10

10 features → $2^{10} - 1$ → NP

Feature Selection :

Supervised ① Filtering methods → Correlation analysis / Mutual Information

② Wrapper methods

Unsupervised ③

1 2 3 4 → 15

Objective function

Accuracy | error

$MI(X_i, Y)$

Rank: $x_1 > x_2 > x_3 \ldots\ldots$

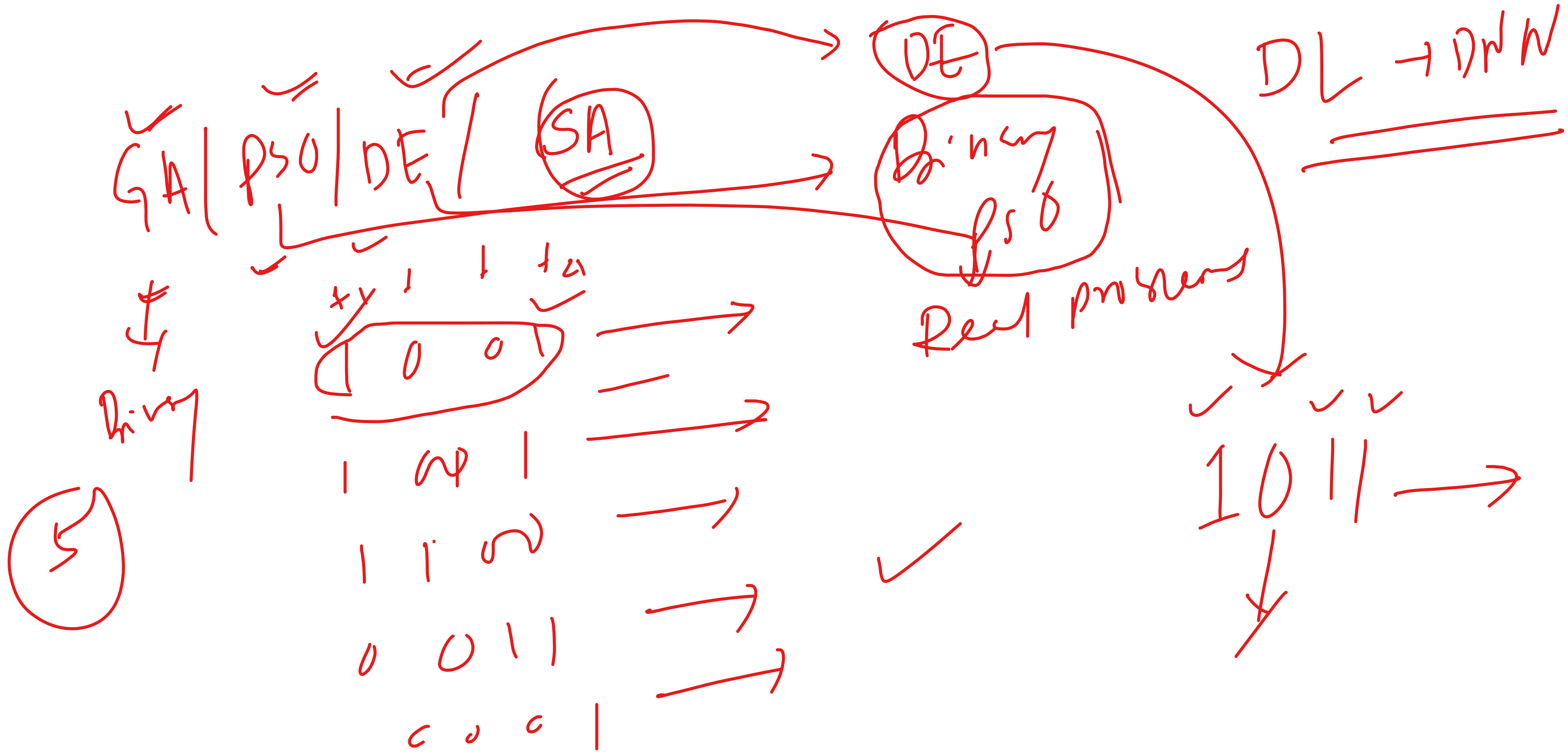
Threshold

$$R(i) = \frac{Cov(x_i, Y)}{\sqrt{V(x_i) \cdot V(Y)}}$$

feature

class label

$Cov(\cdot) = 0$ uncorrelated



GA | PSO | DE | SA
DE
Binary PSO
Real problems
DL → DNN
Binary
5
1 0 0 1
1 0 0 1
1 1 0 0
0 0 1 1
0 0 0 1
1 0 1 1

* Type-I & Type-II Errors:

* p-test, t-test;

End

All the best!